EmoViS: Emoji-Augmented ViSoBERT for Fine-Grained Multi-Label Emotion Recognition in Vietnamese Social Media Text

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Abstract. Fine-grained emotion recognition in Vietnamese social-media text combines informal language, frequent emoji, and a long-tailed, 28-label prediction space. This paper critically evaluates EmoViS, a dual-branch system that fuses a ViSoBERT representation of normalized text with fixed Emoji2Vec features extracted from the original Unicode sequence. On the 2,067-comment ViGoEmotions test split, the three-seed EmoViS system obtains a Macro-F1 of 0.6201 (standard deviation 0.0020), compared with 0.6173 (standard deviation 0.0052) for the weighted-BCE ViSoBERT baseline; this difference is small and not statistically significant. A probability ensemble from the original execution of this configuration reaches a Macro-F1 of 0.6329 and raises mAP from 0.6533 to 0.6875, but this gain is attributable to the complete system, including ensembling and threshold selection. In a sequential ablation, adding the emoji branch recovers 0.0198 Macro-F1 relative to an underperforming asymmetric-loss configuration. In three-seed ablations under weighted BCE, the ensemble Macro-F1 is 0.6297 with textual emoji descriptions, 0.6298 with an added Emoji2Vec branch (paired bootstrap difference +0.0001, p = 0.4891), and 0.6274 with raw Unicode emoji (difference -0.0024, p = 0.7304). Neither alternative differs significantly from textual descriptions, so the branch is best regarded as a component of the full pipeline rather than an independently established source of gain. The separate contributions of asymmetric loss and class-balanced reweighting remain to be factorized.

Keywords: Vietnamese emotion recognition · Multi-label classification · Emoji representation · ViSoBERT · Statistical evaluation

1 Introduction

Vietnamese social-media language combines informal orthography, code switching, slang, and non-verbal symbols. Emotion recognition in this setting is therefore

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not merely polarity classification: the ViGoEmotions benchmark asks a model to assign any subset of 28 labels to each comment [9]. Its authors show that emoji treatment materially affects downstream performance and that preserving raw emoji is the best reported preprocessing choice for ViSoBERT, a language model pretrained specifically on Vietnamese social-media text [9, 15]. This makes an explicit comparison between token-level emoji handling and external emoji features scientifically important.

The original motivation for EmoViS was to preserve an emoji signal in parallel with normalized text. The model uses ViSoBERT for text and Emoji2Vec for a separate fixed feature channel [5]. Emoji2Vec is trained from English Unicode descriptions, however, and therefore cannot by itself justify claims about Vietnamese-specific emoji semantics. Moreover, the first experimental sequence replaced weighted binary cross entropy (BCE) with asymmetric loss (ASL) before adding the emoji branch. Because ASL underperformed the baseline, a statistically significant A1-to-A2 recovery does not prove that the branch beats the stronger A0 baseline. This paper therefore makes those distinctions explicit instead of presenting the sequential recovery as an isolated architectural gain.

The paper makes three evidence-bounded contributions. First, it describes a lightweight dual-branch fusion architecture that retains both ViSoBERT's text representation and an explicit continuous emoji channel. Second, it applies a reproducible evaluation protocol with three seeds, validation-only threshold selection, threshold-independent mean average precision (mAP), paired bootstrap resampling, and multiplicity correction. Third, it reports negative and contradictory results as first-class findings: ASL, class-balanced reweighting, and two contrastive variants do not reliably improve this task, and three-seed controlled ablations under weighted BCE show no statistically significant Macro-F1 difference among textual emoji descriptions, an added Emoji2Vec branch, and raw Unicode emoji. Rather than claiming a new state of the art, the study reports the controlled comparisons needed to interpret the architecture and identifies the remaining factorial comparison.

2 Related Work

2.1 Vietnamese Emotion Recognition

PhoBERT established strong monolingual pretraining for Vietnamese [14], while ViSoBERT adapted XLM-R-style pretraining to Vietnamese social-media text [2, 15]. UIT-VSMEC contains 6,927 social-media sentences annotated with six basic emotions [8]; ViGoEmotions substantially increases granularity and scale to 20,664 comments and 28 output labels (27 emotions plus neutral) [9]. ViGoEmotions evaluates multiple emoji preprocessing policies and reports that raw-emoji preservation performs best for ViSoBERT, which motivates including that policy as a required comparator rather than assuming an external embedding is inherently better.

Table 1. ViGoEmotions statistics used in the experiments.

Split	Comments	Positive labels	Labels/comment	Emoji (%)
Train	16,531	31,545	1.908	25.01
Validation	2,066	3,958	1.916	24.78
Test	2,067	3,942	1.907	26.03
Total	20,664	39,445	1.909	–

2.2 Emoji Representation

Emoji provide affective and pragmatic cues that plain text can under-specify. Prior resources include an emoji sentiment ranking derived from large-scale tweets [11], Emoji2Vec embeddings learned from Unicode descriptions [5], and DeepMoji representations pretrained on emoji prediction over 1.2 billion tweets [6]. EmoViS uses Emoji2Vec because it is compact and easy to reproduce, not because it captures Vietnamese community semantics. The distinction matters: a fixed English-description embedding supplies a cross-lingual semantic prior, whereas raw emoji tokens processed by ViSoBERT may better preserve usage patterns learned from Vietnamese social media.

2.3 Long-Tail and Contrastive Learning

Weighted BCE remains a strong baseline for multi-label imbalance. Alternatives include focal loss [13], label-distribution-aware margins [1], class-balanced weighting [3], distribution-balanced loss [17], and ASL [16]. For representation regularization, R-Drop penalizes disagreement between stochastic forward passes [12]. TACO was developed for fine-grained single-label emotion classification [7], whereas MulSupCon weights positive relationships according to label overlap in multi-label data [18]. Their assumptions do not transfer automatically to sparse, overlapping Vietnamese emotion labels.

3 Dataset and Task

ViGoEmotions contains 20,664 public social-media comments split 8:1:1 into train, validation, and test sets [9]. Each sample may receive several of 28 outputs; the average is about 1.91 positive labels per comment. Table 1 reports the exact split statistics used in this study. The training distribution is long-tailed: the most frequent emotion has 2,868 training positives and the least frequent has 635.

A raw-source audit found one repeated label identifier in one test row, while the parsed binary target matrix and its per-class supports still contain exactly 3,942 positives. The headline ensemble score is reproduced by re-executing the original notebook. Because it was produced in that execution rather than in

the rerun used for the three-run summaries, this paper distinguishes *original-notebook outputs* from independently rerun summaries and does not treat them as a matched pair.

4 Method

4.1 Text Baseline

All main configurations use the same ViSoBERT encoder, preprocessing, maximum length, optimization schedule, and *masked mean pooling*. If $\mathcal{T}$ denotes non-padding token positions, the text representation is

$$\mathbf{h}_t = \frac{1}{|\mathcal{T}|} \sum_{j \in \mathcal{T}} \mathbf{h}_j \in \mathbb{R}^{768}. \tag{1}$$

The weighted-BCE baseline uses one positive weight per class,

$$w_k = \frac{N - n_k}{n_k}, \tag{2}$$

where N is the number of training samples and n_k the positives for class k.

4.2 Explicit Emoji Branch

The data pipeline retains both the original string and its normalized text. Emoji are extracted from the original Unicode sequence before normalization. For the set E of in-vocabulary emoji, fixed 300-dimensional Emoji2Vec vectors are averaged:

$$\bar{\mathbf{v}}_e = \begin{cases} |E|^{-1} \sum_{e \in E} \mathbf{v}_e, & |E| > 0, \\ \mathbf{0}, & |E| = 0. \end{cases} \tag{3}$$

The result is projected using

$$\mathbf{h}_e = \mathrm{LayerNorm}(\mathrm{GELU}(W_e \bar{\mathbf{v}}_e + \mathbf{b}_e)) \in \mathbb{R}^{768}. \tag{4}$$

Because the projection contains a bias, an emoji-free sample does not necessarily map to the all-zero hidden vector. The two branches are concatenated and passed through a 768-unit GELU fusion layer, dropout, and a 28-output linear classifier:

$$\hat{\mathbf{y}} = \sigma(W_2 \, \mathrm{Dropout}(\mathrm{GELU}(W_1[\mathbf{h}_t; \mathbf{h}_e] + \mathbf{b}_1)) + \mathbf{b}_2) \, . \tag{5}$$

This architecture adds an explicit emoji prior; it is not claimed to model Vietnamese-specific emoji semantics. Figure 1 illustrates the complete EmoViS pipeline, including the normalized text branch, the raw-emoji branch, and the final fusion network.

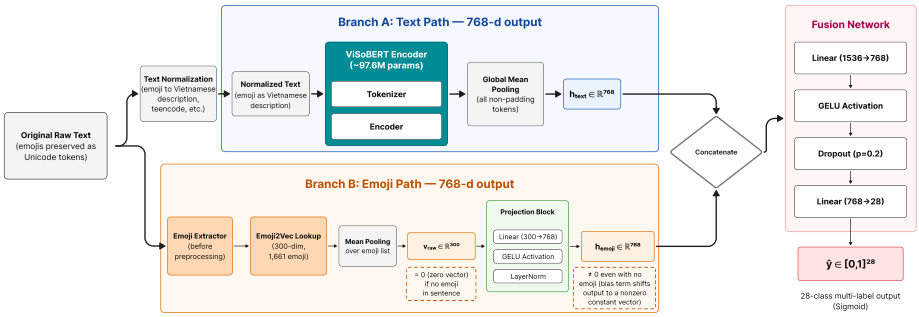


Fig. 1. Overview of the EmoViS architecture. The upper branch encodes normalized Vietnamese social-media text with ViSoBERT and masked mean pooling to obtain a 768-dimensional text representation. The lower branch extracts raw Unicode emoji before normalization, looks up fixed 300-dimensional Emoji2Vec embeddings, averages them, and projects the result into a 768-dimensional emoji representation. The two vectors are concatenated and passed through a fusion network with a linear layer, GELU, dropout, and a final 28-way sigmoid classifier for multi-label emotion prediction.

4.3 Investigated Objectives

ASL uses separate focusing exponents for positive and negative targets and clips easy negative probabilities [16]. Experiments set $\gamma^+ = 0$, $\gamma^- = 4$, and clipping margin 0.05. Class-balanced (CB) reweighting uses the effective number of samples [3],

$$w_k^{\mathrm{CB}} = \frac{1-\beta}{1-\beta^{n_k}}, \qquad \beta = 0.999, \tag{6}$$

normalized to unit mean. This is loss reweighting, not resampling. Since the resulting weights are nearly uniform over the observed counts and partly overlap with existing positive weighting, CB is treated as an investigated option rather than part of a validated contribution.

4.4 Ensembling

For seeds $\{42, 1, 7\}$, the final probability vector is

$$\hat{\mathbf{p}}_{\mathrm{ens}} = \frac{1}{3} \sum_{s \in \{42,1,7\}} \hat{\mathbf{p}}_s. \tag{7}$$

Class thresholds are selected only on validation predictions, after probability averaging. This improves the complete prediction pipeline but does not isolate the effect of the emoji branch.

4.5 Design Rationale and Complexity

The model uses feature-level concatenation instead of logit averaging because the intended question is whether an explicit emoji representation supplies information

Table 2. Shared training configuration for all main runs.

Setting	Value
Encoder	`uitnlp/visobert` (≈97.6M parameters)
Optimizer	AdamW, weight decay 0.01
Learning rate	5×10^{-5}, 1 warmup epoch, linear decay
Batch size / max length	32 / 160 tokens
Dropout / max epochs	0.2 / 12 (early-stopping patience 3)
Seeds	42, 1, 7
Emoji features	Emoji2Vec, 300-d, 1,661 emoji, fixed
ASL	$\gamma^+ = 0$, $\gamma^- = 4$, clip 0.05
Threshold search	0.05–0.94, step 0.01, clipped to $[0.15, 0.85]$
Bootstrap	10,000 paired resamples, test $n = 2{,}067$
Hardware	2× NVIDIA T4

that the language representation can use jointly. Mean pooling in the emoji branch makes inference independent of emoji count and introduces only a projection and a small fusion multilayer perceptron beyond ViSoBERT. The branch therefore adds roughly 1.4 million trainable parameters: about 0.23 million for the 300×768 projection and about 1.18 million for the 1536×768 fusion matrix, plus biases and normalization parameters. The fixed Emoji2Vec lookup itself is not updated.

This compact design has two limitations. First, averaging discards emoji order, repetition strength, skin-tone modifiers, and interactions between emoji and nearby words. Second, an out-of-vocabulary emoji is represented as zero before a biased projection; its downstream representation can consequently resemble the learned default for emoji-free text. A production implementation should include an explicit "emoji present" mask and Unicode-sequence-aware tokenization. These limitations motivate treating EmoViS as an interpretable feature-fusion baseline rather than a complete multimodal architecture.

5 Experimental Protocol

All main runs use `uitnlp/visobert` (approximately 97.6 million parameters), AdamW with learning rate 5×10^{-5} and weight decay 0.01, batch size 32, maximum sequence length 160, dropout 0.2, and at most 12 epochs. A one-epoch linear warmup precedes linear decay; early stopping uses patience 3. The historical runs used two NVIDIA T4 GPUs. Table 2 summarizes the shared configuration.

The primary metric is Macro-F1 with class-specific thresholds selected on validation data over 0.05–0.94 in increments of 0.01 and clipped to $[0.15, 0.85]$. Classes with fewer than ten validation positives retain 0.5. Macro-F1 at threshold 0.5, Micro-F1, and threshold-independent mAP are also reported. Since threshold optimization can inflate apparent gains, fixed-threshold and mAP values remain visible throughout.

Each main configuration is independently trained with three seeds. Paired bootstrap resampling uses 10,000 matched resamples of the 2,067 test examples [4,

Table 3. Controlled sequential configurations. Each row changes one item from the preceding row.

ID	Name	Change
A0	Baseline	ViSoBERT, masked mean pooling, weighted BCE
A1	ASL	Replace weighted BCE with ASL
A2	EmoViS	Add the Emoji2Vec branch to A1
A3	EmoViS+CB	Add effective-number reweighting to A2

10]. Thresholds remain fixed during resampling. For class-level tests, Holm–Bonferroni correction controls multiplicity. Bootstrap differences computed from probability-averaged predictions are not algebraically equal to differences between the mean of three independently thresholded F1 scores; the tables label these quantities separately.

For class k, precision and recall are computed from TP_k, FP_k, and FN_k, and the class F1 is

$$F1_k = \frac{2TP_k}{2TP_k + FP_k + FN_k}. \tag{8}$$

Macro-F1 is $K^{-1}\sum_{k=1}^{K} F1_k$ with $K = 28$, assigning equal weight to frequent and rare labels. Micro-F1 aggregates all class decisions before computing the ratio and therefore emphasizes common labels. Average precision integrates ranking quality for one class without choosing a decision threshold; mAP is its macro-average over labels. Reporting all three views helps expose models that appear strong only because of threshold tuning or majority labels.

For a comparison between systems a and b, each bootstrap draw samples test indices with replacement and computes $d^{(r)} = \mathrm{MF1}_b^{(r)} - \mathrm{MF1}_a^{(r)}$. The percentile interval uses the 2.5th and 97.5th percentiles of $\{d^{(r)}\}_{r=1}^{10000}$. The one-sided empirical p-value estimates the fraction of draws with $d^{(r)} \leq 0$. This procedure captures test-sample uncertainty conditional on the trained checkpoints; it does not replace uncertainty across training seeds.

6 Results

6.1 Main System Results

Table 3 defines configurations A0–A3. Table 4 separates independently trained model statistics from the single probability-ensemble output. A3 is only 0.0028 above A0 in mean Macro-F1, smaller than run-to-run variation and not established as significant by the available controlled evidence. The ensemble reaches 0.6329 Macro-F1 and 0.6875 mAP. The latter improvement is useful because mAP is independent of the chosen classification thresholds. Nevertheless, this result supports the *complete ensembled system*, not the emoji branch in isolation. This ensemble comes from the original notebook execution, whose recorded per-seed Macro-F1 values for the same configuration were 0.6316, 0.6225, and 0.6319 for

Table 4. Main test results. MF1-opt uses validation-selected thresholds; MF1-0.5 uses a fixed threshold.

System	MF1-opt	MF1-0.5	Micro-F1	mAP
ViSoBERT, original report [9]	0.6150	–	–	–
A0, three runs	0.6173 ± 0.0052	0.6147 ± 0.0063	0.6378 ± 0.0030	0.6533 ± 0.0017
A3, three runs	0.6201 ± 0.0020	0.5857 ± 0.0008	0.6316 ± 0.0021	0.6618 ± 0.0011
A3 probability ensemble (historical run)	**0.6329**	**0.6214**	**0.6550**	**0.6875**

Table 5. Sequential ablation. Bootstrap values are paired comparisons of three-seed averaged probabilities from stored predictions, each relative to the preceding row; they are not differences between the three-run means.

Configuration	MF1-opt (mean±sd)	Mean change	Bootstrap change	One-sided p
A0: weighted BCE	0.6173 ± 0.0052	–	–	–
A1: +ASL	0.6115 ± 0.0040	-0.0058	-0.0399	1.000
A2: +emoji branch	0.6138 ± 0.0023	$+0.0023$	$+0.0198$	< 0.0001
A3: +CB	0.6201 ± 0.0020	$+0.0063$	-0.0019	0.757

seeds 42, 1, and 7; these are higher than the rerun A3 seed scores in Table 6, so the ensemble and the rerun three-run statistics should not be read as one matched experiment.

The per-seed MF1-opt values are A0: 0.6133/0.6231/0.6155; A1: 0.6070/0.6128/0.6147; A2: 0.6165/0.6126/0.6122; and A3: 0.6190/0.6188/0.6224 for seeds 42/1/7. Thus, the fused ASL model A2 has a lower three-seed mean than weighted-BCE A0. This directly prevents a claim that the emoji branch beats the baseline under the current sequence.

6.2 Sequential Ablation

Table 5 separates two kinds of change. The "mean change" column compares the means of independently thresholded runs; the bootstrap column compares specified probability objects under fixed validation-derived thresholds. Replacing weighted BCE with ASL reduces mean MF1-opt by 0.0058. Adding the emoji branch then raises the paired bootstrap statistic by 0.0198 relative to the degraded ASL model. This is evidence that the branch can recover information lost in A1, but it is not evidence of improvement over A0.

The apparent A3 increase in the mean column and decrease in the bootstrap column are compatible because the two columns summarize different objects (means of independently thresholded runs versus probability-averaged predictions) under a nonlinear, threshold-dependent metric. They should not be merged into a single causal claim. The corresponding component test, with weighted BCE held fixed across three seeds, is reported in Section 6.4.

Table 6 exposes every run instead of relying only on mean and standard deviation. A0 has the highest single seed among A0–A3 at seed 1, while A3 is more stable and has its highest score at seed 7. The crossing rank order illustrates

Table 6. Per-seed Macro-F1 with validation-selected thresholds.

System	Seed 42	Seed 1	Seed 7	Mean	Std.
A0	0.6133	**0.6231**	0.6155	0.6173	0.0052
A1	0.6070	0.6128	0.6147	0.6115	0.0040
A2	0.6165	0.6126	0.6122	0.6138	0.0023
A3	0.6190	0.6188	0.6224	**0.6201**	**0.0020**

Table 7. Supplementary single-run loss comparison on the fused architecture (seed 42).

Objective	MF1-opt	MF1-0.5	F1 on eight weak labels
BCE (unweighted)	0.6305	0.6096	0.4870
Weighted BCE	**0.6435**	0.6168	0.4882
Focal loss	0.6349	**0.6165**	0.5059
LDAM+DRW	0.6295	0.6147	0.4971
Distribution-balanced	0.6272	0.6131	**0.5105**
ZLPR	0.6350	0.6143	0.4875
ASL	0.6369	**0.6165**	0.5047

why a claim based on one seed would be fragile. It also explains why a probability ensemble can exceed the arithmetic mean: averaging scores is not equivalent to averaging probabilities and then applying nonlinear, class-specific thresholds.

6.3 Supplementary Loss Study

A separate seed-42 experiment applies several losses to the fused architecture (Table 7). These runs come from a separate execution, and their absolute values are not directly comparable with the matched three-seed rows in Table 6: the ASL row here (0.6369) is higher than the matched A2 seed-42 score of 0.6165. Within this table, weighted BCE is the strongest observed single run at 0.6435 MF1-opt, exceeding ASL at 0.6369 and the reported A3 ensemble at 0.6329. Because the runs are not matched across three seeds, this cannot establish population-level superiority; it does, however, indicate that weighted BCE is a competitive alternative to ASL, which Section 6.4 evaluates under three seeds. In that rerun, the fused weighted-BCE configuration (C1) has a seed-mean of 0.6207 ± 0.0015, so the 0.6435 score should be read as a single-run observation rather than as the expected performance of the configuration.

6.4 Controlled Weighted-BCE Ablation

To test the emoji branch against the stronger weighted-BCE baseline directly, a three-seed controlled ablation (seeds 42, 1, and 7) was executed separately with a pinned software environment (`transformers` 4.52.4). Configuration C0 uses weighted-BCE ViSoBERT with emoji converted to textual descriptions; C1 adds the fixed Emoji2Vec branch to C0; and C2 keeps weighted BCE and no external

Table 8. Controlled weighted-BCE ablation over three seeds. Bootstrap Δ and one-sided p are relative to C0 (ns: $p \geq 0.05$). The EmoViS-Ens row is the full ASL and class-balanced system from the earlier execution, shown for reference; † denotes a difference of ensemble scores without a paired test.

System	Emoji handling	Seed-mean MF1-opt	Ens. MF1-opt	Ens. Micro-F1	Ens. mAP	Bootstrap Δ	p
C0: weighted BCE	Text descriptions	0.6216 ± 0.0015	0.6297	0.6394	0.6672	–	–
C1: +Emoji2Vec branch	Descriptions + Emoji2Vec	0.6207 ± 0.0015	0.6298	0.6422	0.6716	+0.0001	0.4891 (ns)
C2: raw Unicode	Raw emoji in text	0.6210 ± 0.0009	0.6274	0.6434	0.6703	−0.0024	0.7304 (ns)
EmoViS-Ens (ASL+CB)	Descriptions + Emoji2Vec	–	0.6329	–	–	+0.0032†	–

branch but passes raw Unicode emoji to ViSoBERT. Thresholds are selected on validation data only, ensembles average the three seed-level probability outputs, and bootstrap comparisons are paired and computed relative to C0 (Table 8). Because this is a separate execution, absolute values are compared within Table 8 and are not pooled with Tables 4 and 6.

Under an identical objective, neither alternative differs significantly from C0 in Macro-F1. Adding the Emoji2Vec branch changes the ensemble score by +0.0001 ($p = 0.4891$), and keeping raw Unicode emoji changes it by −0.0024 ($p = 0.7304$). Both emoji-aware variants show higher ensemble Micro-F1 and mAP than C0, but these secondary differences were not tested and are reported descriptively. The original benchmark reports raw-emoji preservation as its best preprocessing policy for ViSoBERT [9]; in the present pipeline, raw Unicode and textual descriptions are statistically indistinguishable on Macro-F1, so the comparison neither reproduces nor contradicts that ranking at the available power.

For reference, the full EmoViS-Ens ensemble (0.6329) is 0.0032 above the C0 ensemble. Since it originates from the earlier execution and no paired test is available, this difference is descriptive. The controlled design also does not separate asymmetric loss, class-balanced reweighting, and ensembling settings from the other differences between the full system and C0, so their individual contributions remain to be factorized.

6.5 Regularization and Contrastive Results

Starting from a historical seed-42 fused ASL model reported at 0.6236 in the original notebook run (the matched three-seed protocol gives 0.6165 for A2 at seed 42, Table 6; the two values come from different executions and are not interchangeable), R-Drop reaches 0.6295 for $\alpha = 1.0$ and 0.6260 for $\alpha = 0.3$. TACO variants range from 0.5881 to 0.6113, and MulSupCon obtains 0.5708. These are exploratory single-run results only. A diagnostic analysis reported that 15.28% of cluster-negative pairs shared an original label even though the stated pairing rule should make that impossible. Because this indicates either incomplete cluster coverage or a mismatch in the measurement script, the TACO result is reported as an unresolved implementation finding, not a reliable negative conclusion about the published method.

Table 9. Per-class F1 for six difficult labels. Asterisk marks significance after Holm–Bonferroni correction.

Label	Support	A0	Ensemble	Difference
Disapproval	112	0.3445	0.4678	+0.1233*
Neutral	116	0.4171	0.3810	−0.0361
Disappointment	197	0.4673	0.5047	+0.0374
Nervousness	96	0.4737	0.5143	+0.0406
Confusion	84	0.4972	0.4757	−0.0215
Realization	95	0.5137	0.4862	−0.0275

6.6 Per-Class Behavior

Among six difficult classes, the ensemble improves disapproval, disappointment, and nervousness, while neutral, realization, and confusion regress (Table 9). Only disapproval retains statistical significance after Holm–Bonferroni correction: bootstrap mean difference 0.1121, 95% CI $[0.0475, 0.1789]$, adjusted $p = 0.0024$. Point-estimate differences in the table and bootstrap mean differences are deliberately kept distinct. The result does not show that emoji alone causes the disapproval gain, because the tested ensemble also changes loss weighting and averages checkpoints.

6.7 Qualitative Error Audit

A0 and the historical ensemble disagree on 1,217 of 2,067 test comments (58.9%). Two authors manually inspected 20 anonymized disagreements drawn from four high-disagreement classes, five comments per class. The non-exclusive coding categories were eight cases with fewer false positives, four recovered labels, seven cases where both systems erred differently, and five new false positives. Since categories can overlap and the sample is small, these counts are diagnostic rather than inferential.

The audit suggests three recurring error sources. First, comments whose literal text is neutral but whose emoji is affective may benefit from an explicit emoji signal, although the controlled comparisons in Section 6.4 show no statistically significant aggregate Macro-F1 benefit from the external branch. Second, cognitive labels such as realization and confusion often require discourse context outside a single comment. Third, multi-label boundary ambiguity creates plausible partial predictions: a model may identify annoyance but miss disapproval, or identify sadness but miss disappointment. Future qualitative work should use a preregistered coding guide, blinded raters, and agreement statistics rather than post-hoc inspection by model developers.

7 Discussion

7.1 What the Evidence Supports

The clearest system-level result is that the historical three-seed A3 probability ensemble, followed by validation-only thresholding, scores higher than the reproduced A0 summary on Macro-F1 and mAP, although the two were not produced in the same execution. The effect cannot be assigned solely to emoji fusion because A3 changes loss, fusion, reweighting, and ensembling, and its pre-ensemble mean is close to A0. The sequential A1-to-A2 comparison demonstrates recovery relative to ASL, not superiority over weighted BCE, and the controlled weighted-BCE ablation (Section 6.4) likewise finds no significant Macro-F1 change from the external branch or from raw Unicode emoji. Taken together, the evidence supports the full pipeline as a strong configuration on ViGoEmotions while leaving the attribution among its components open.

The fixed-threshold decrease under ASL should not be explained as a simple uniform downward probability shift: diagnostic class means increased for several labels even as fewer nervousness predictions crossed 0.5. A more accurate interpretation is that ASL changes probability calibration and ranking in a class-dependent manner. Reliability diagrams, expected calibration error, and class-wise score distributions are needed before making a mechanistic calibration claim.

7.2 Remaining Comparisons

The two comparisons required by the sequential design, namely weighted BCE with the external branch toggled and raw emoji preserved in the ViSoBERT input, are reported in Section 6.4. What remains is to factorize the other differences between the full pipeline and C0: asymmetric loss, class-balanced reweighting, and ensembling settings. A design over the external branch (off versus on), loss (weighted BCE versus ASL), and class-balanced reweighting (off versus on), repeated for the same seeds, would separate these effects.

A second Vietnamese corpus would strengthen external validity, but existing corpora differ in taxonomy and granularity. UIT-VSMEC is single-label with six emotions, so cross-dataset evidence should either reconfigure the output head and report a separate task or map only semantically compatible labels; it should not be portrayed as direct 28-label replication. The present single-benchmark results are therefore evidence about ViGoEmotions rather than a universal conclusion about Vietnamese emotion recognition.

7.3 Operational Relevance

The work is relevant to intelligent social-media analytics and edge/cloud information-processing systems. A deployment can use fine-grained emotion outputs for aggregate monitoring, moderation triage, or adaptive human–machine interfaces. Such use requires calibration, drift monitoring, privacy safeguards, and human

review because minority-label errors and platform-specific language can produce disproportionate harms.

A practical serving stack can compute Emoji2Vec features on-device or at an edge gateway because lookup and averaging are inexpensive, while ViSoBERT inference can run on an edge accelerator or cloud service. The extra branch has negligible sequence-length cost, but the encoder remains the dominant latency and energy component. For live streams, the system should monitor label prevalence, emoji out-of-vocabulary rate, confidence distributions, and language drift. Alerts should be based on aggregates rather than individual emotional profiling unless users have given informed consent.

7.4 Threats to Validity

Internal validity. The sequential ablation changes the loss before adding the branch, so it does not by itself identify the branch effect against A0; the controlled weighted-BCE ablation in Section 6.4 addresses that question directly. The headline ensemble is reproducible by re-executing the original notebook, but it was not produced in the same execution as the three-run summaries, so the two are not a matched pair. The contributions of asymmetric loss, class-balanced reweighting, and ensembling to the headline score also remain unfactorized; these issues are addressed by narrowing claims, and only rerunning the corresponding controlled cells can remove the threat.

Construct validity. Macro-F1 depends on validation-tuned thresholds and may reward threshold search on rare labels. Fixed-threshold Macro-F1 and mAP reduce, but do not eliminate, this concern. Emotion labels are also constructs with overlap and annotator subjectivity; a prediction that differs from the stored label is not necessarily pragmatically unreasonable.

External validity. One dataset cannot establish robustness across platforms, dialects, time periods, or annotation taxonomies. Emoji2Vec's English-description origin further limits cultural interpretation. Any transfer claim should be evaluated on a separate corpus without reusing test labels or thresholds.

Conclusion validity. Three seeds are better than one but still provide a weak estimate of training variance. Test-set bootstrapping measures sample uncertainty conditional on fitted models, not optimizer uncertainty. Multiple exploratory losses and regularizers also create researcher degrees of freedom; future work should preregister the primary comparison and correction family.

8 Limitations, Ethics, and Reproducibility

The study is limited to one 28-label benchmark. Emoji2Vec covers 1,661 emoji and is derived from English descriptions, so its coverage and cultural semantics are imperfect. The controlled weighted-BCE ablations use three seeds on one

dataset, and the factorization of asymmetric loss and class-balanced reweighting has not yet been run. Several supplementary comparisons are single-run, and the TACO pair-construction diagnostic remains unresolved. Validation-tuned thresholds for rare classes can also overfit small positive counts.

Only the public ViGoEmotions data are used. Examples should remain anonymized, and downstream systems should not infer sensitive traits or make high-stakes decisions from emotion predictions alone. The repository should release configuration files, seeds, package versions, target-matrix checksums, validation thresholds, and test probability arrays for every reported run.

9 Conclusion

This study finds that the current EmoViS evidence supports an ensembled system result, not a clean claim that Emoji2Vec fusion outperforms weighted-BCE ViSoBERT. The three-run A3 model obtains 0.6201 ± 0.0020 Macro-F1 versus 0.6173 ± 0.0052 for A0, while the historical probability ensemble reaches 0.6329 and improves mAP to 0.6875. The sequential emoji addition recovers performance from an underperforming ASL configuration, and three-seed weighted-BCE ablations show no statistically significant Macro-F1 difference among textual emoji descriptions (ensemble 0.6297), an added Emoji2Vec branch (0.6298), and raw Unicode emoji (0.6274). The next step is therefore a preregistered, multi-seed factorial comparison of the external branch, asymmetric loss, and class-balanced reweighting, accompanied by complete probability artifacts and an independent second-dataset evaluation.

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